H2 CFD Mesh-Independence and Verification Study

L7 Stenosed Lumen | OENG1168 Engineering Capstone Project Part B

Prepared by Harshita Bhandari
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Study status: CFD simulation matrix complete

1. Purpose and Scope

H2 was completed as a three-level mesh-sensitivity and CFD verification study for the L7 stenosed lumen. The aim was to determine how the predicted pressure drop and wall shear stress change with mesh refinement, while also checking numerical convergence and mass conservation at two inlet-flow conditions.

The complete 98.2 mm lumen geometry was retained. The lesion-containing 25 mm analysis segment extends from z = 36.6 mm to z = 61.6 mm, with 36.6 mm straight inlet and outlet extensions. The STEP geometry was supplied in millimetres and scaled to metres in Fluent.

2. CFD Setup

Item	H2 setting
Solver	Pressure-based, steady
Flow model	Laminar
Fluid	Blood, Carreau non-Newtonian viscosity
Carreau parameters	λ = 3.313 s; n = 0.3568; a = 2; μ0 = 0.056 Pa·s; μ∞ = 0.00345 Pa·s
Density	1050 kg/m³
Wall	No-slip
Outlet	0 Pa gauge pressure
Inlet	Parabolic profile, R = 0.00122 m
Mean velocities	0.10 m/s and 0.26 m/s
Segment planes	z = 0.0366 m and z = 0.0616 m

Parabolic inlet expression:

$u_z = U_{max} [1 - (x^2 + y^2)/R^2]$, with $U_{max} = 2U_{mean}$.

3. Mesh Levels and Quality

Mesh	Cells	Minimum orthogonal quality	Status
Coarse	20,120	0.201	Accepted
Medium	90,420	0.430	Accepted
Fine	221,409	0.469	Accepted

Using h ∝ N^(-1/3), the local refinement ratios are r21 = 1.348 (medium/fine) and r32 = 1.650 (coarse/medium). Both exceed the requested 1.3 refinement-ratio threshold.

4. Final H2 Results

4.1 Mean inlet velocity = 0.10 m/s

Mesh	Cells	Full ΔP [Pa]	25 mm segment ΔP [Pa]	Max WSS [Pa]
Coarse	20,120	341.737	155.902	17.590
Medium	90,420	322.093	147.753	17.506
Fine	221,409	308.534	140.177	17.202

4.2 Mean inlet velocity = 0.26 m/s

Mesh	Cells	Full ΔP [Pa]	25 mm segment ΔP [Pa]	Max WSS [Pa]
Coarse	20,120	1036.714	590.176	58.158
Medium	90,420	988.640	578.138	62.218
Fine	221,409	924.755	561.069	61.039

All six CFD cases were solved with the same governing setup and post-processing definitions. The pressure monitors reached stable plateaus and the final mass-flow imbalance was negligible in every case.

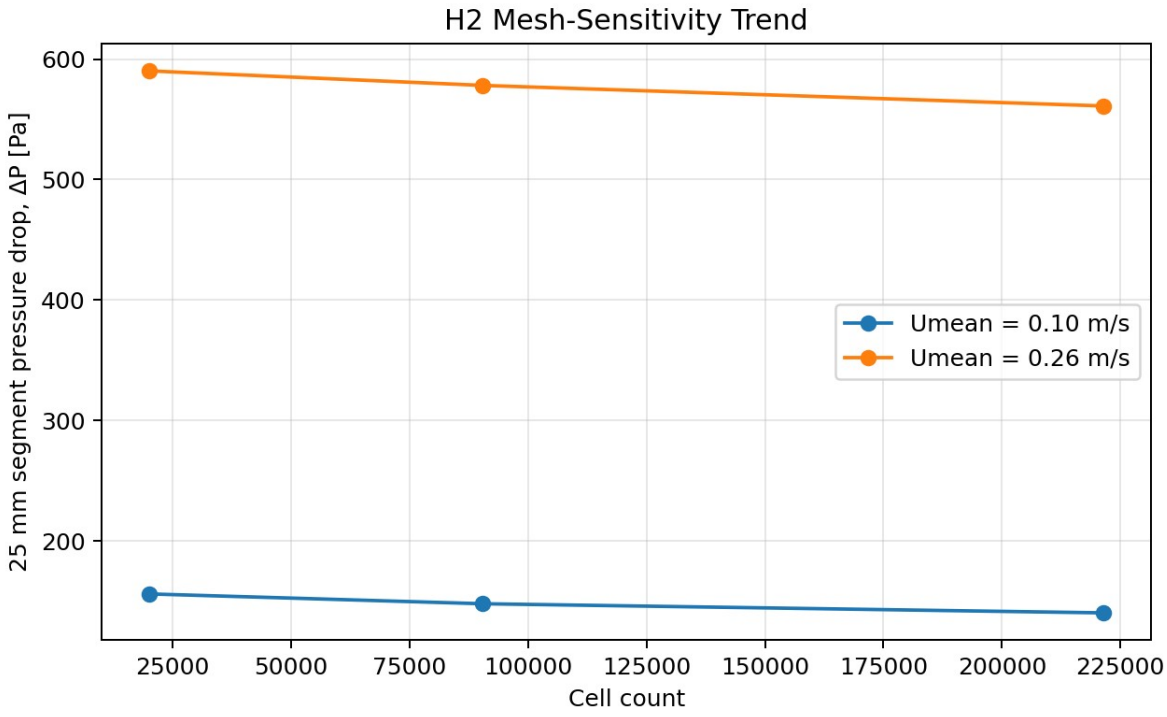


Figure 1. Mesh-sensitivity of the 25 mm lesion-segment pressure drop at both inlet velocities.

5. Numerical Convergence and Result Evidence

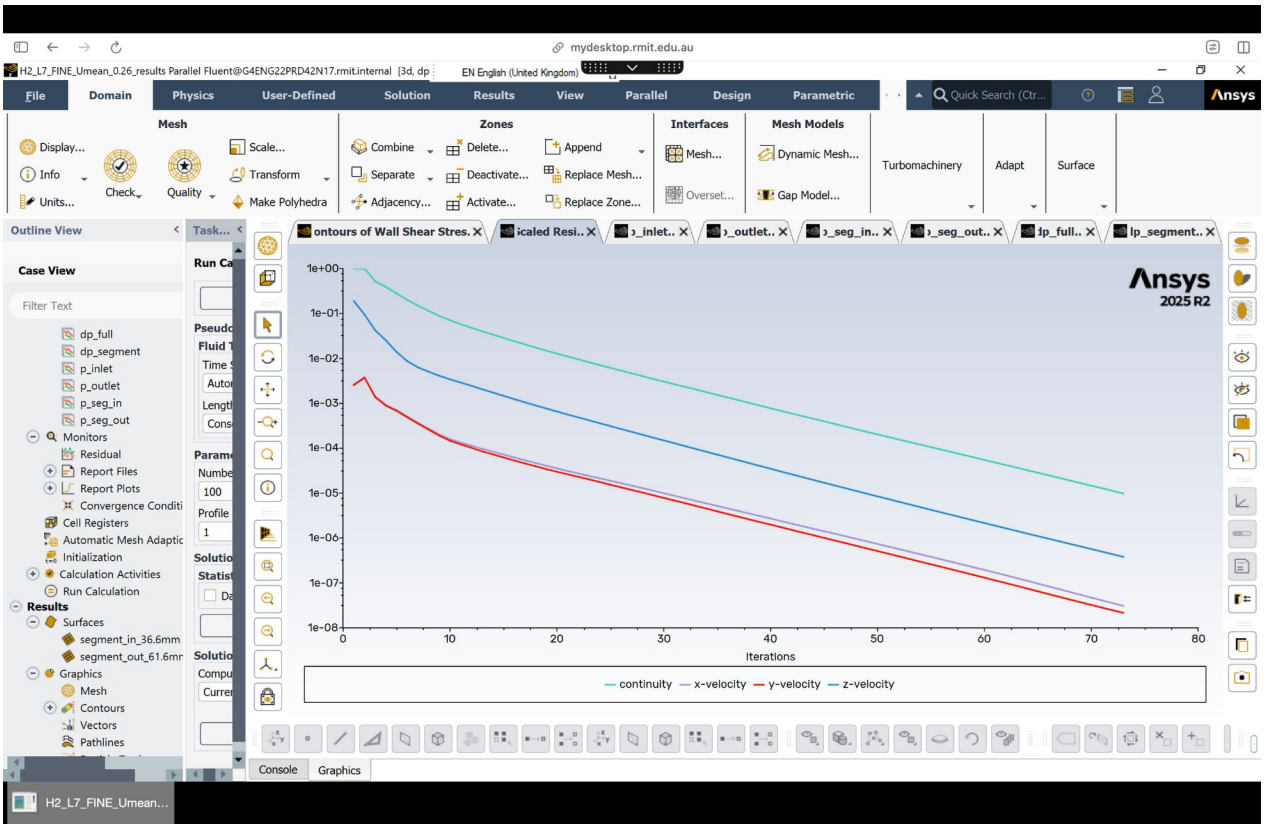


Figure 2. Fine-mesh 0.26 m/s residual history showing smooth residual reduction.

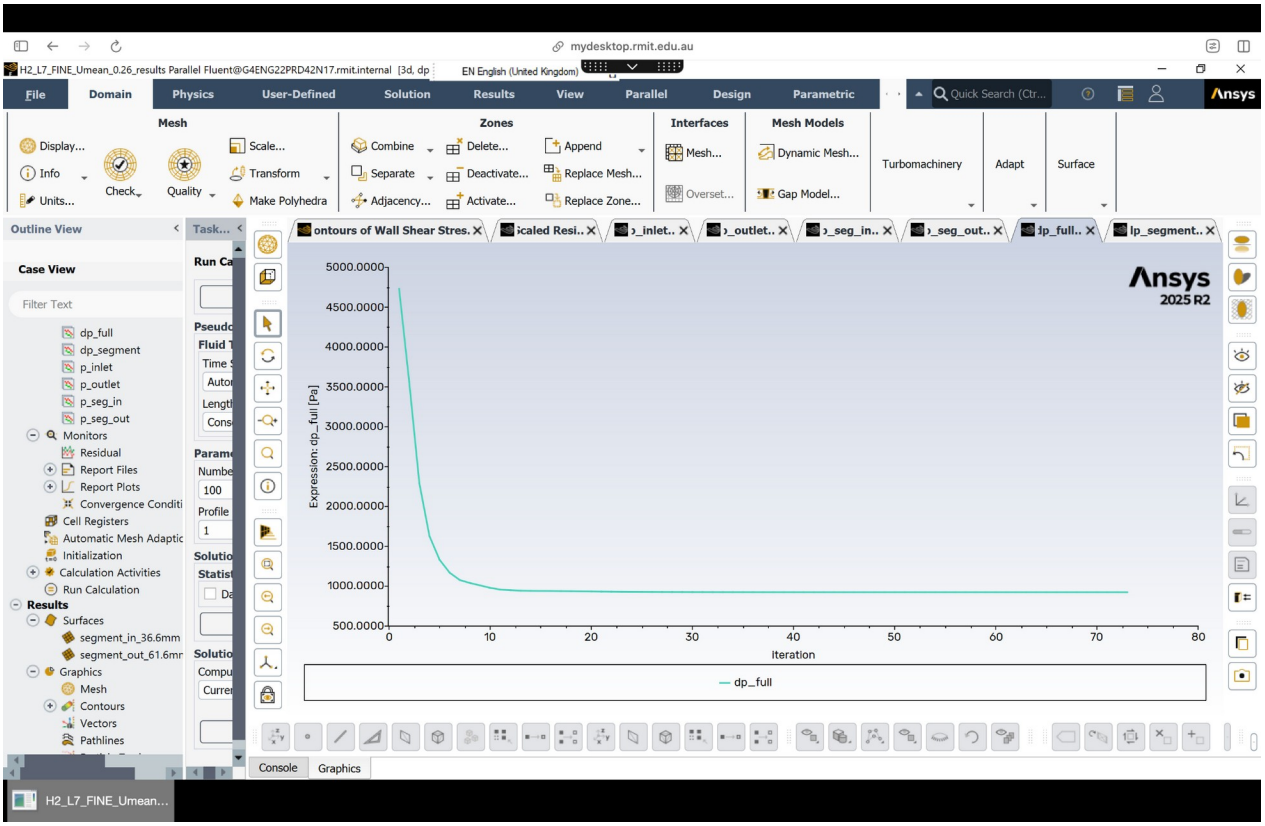


Figure 3. Fine-mesh 0.26 m/s full-domain pressure-drop monitor reaching a stable plateau.

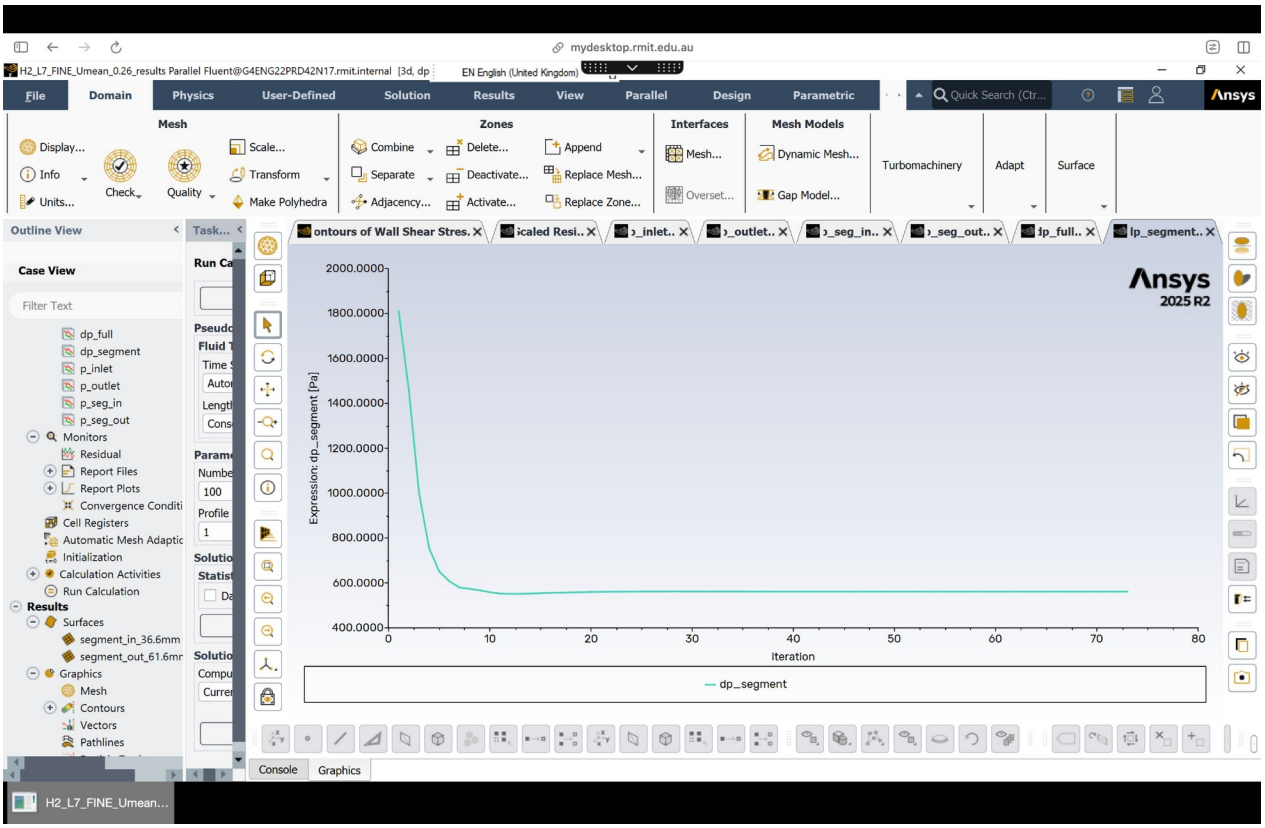


Figure 4. Fine-mesh 0.26 m/s 25 mm segment pressure-drop monitor reaching a stable plateau.

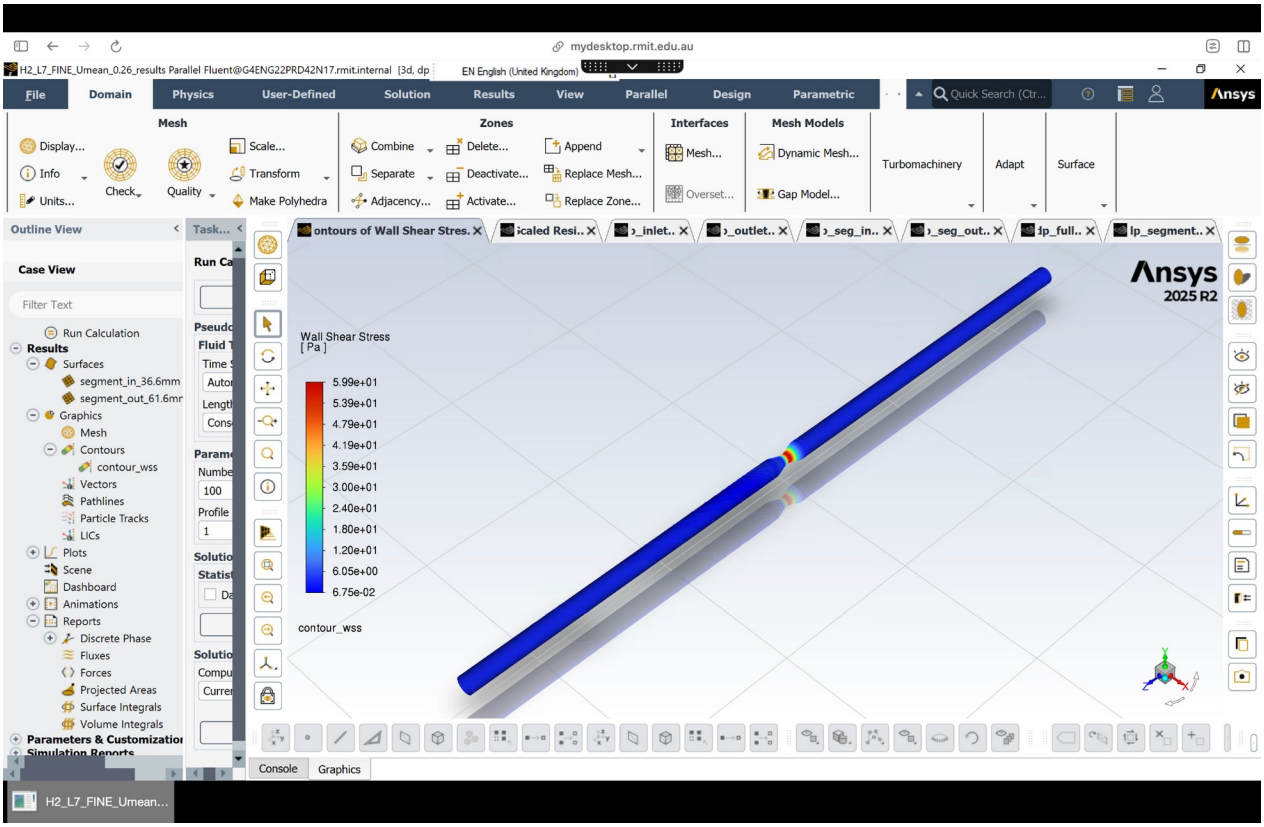


Figure 5. Fine-mesh wall shear stress contour, with the highest WSS concentrated around the stenosis/throat region.

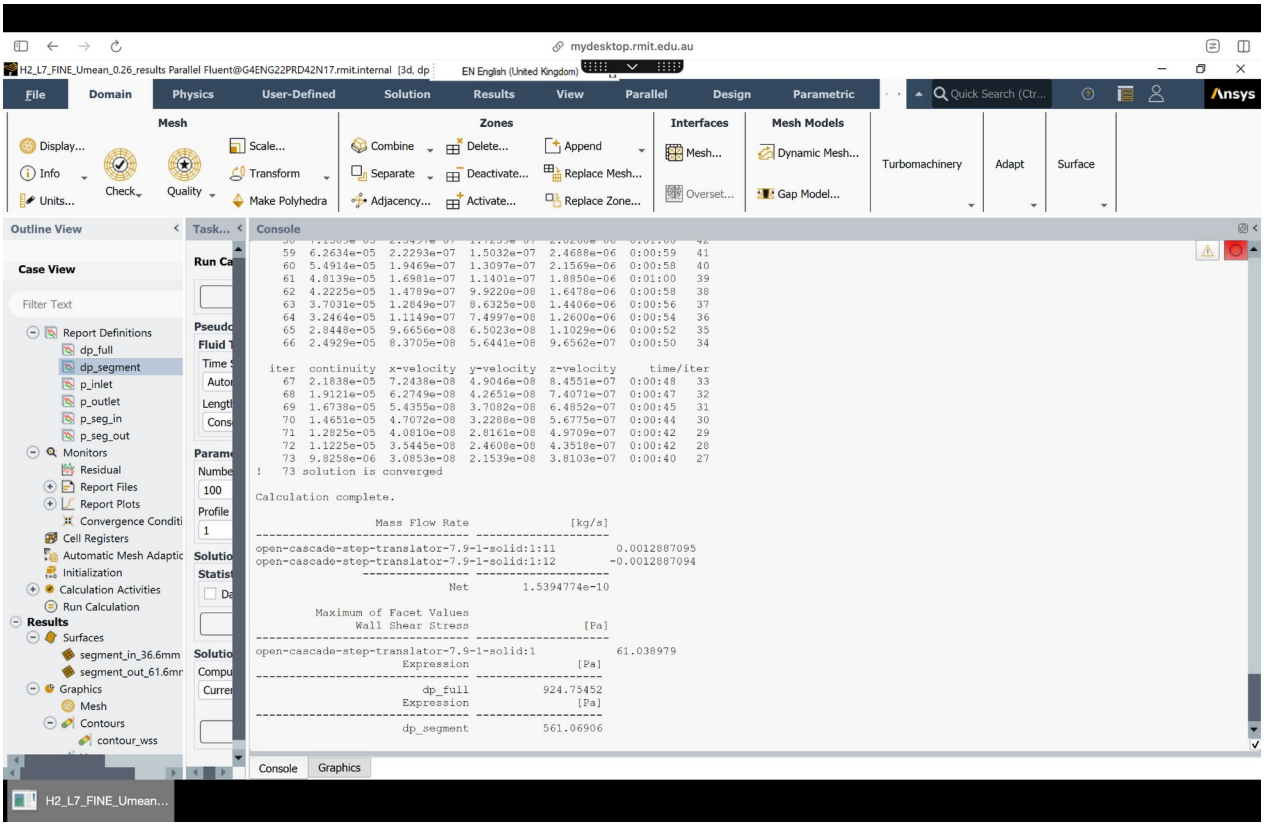


Figure 6. Fine-mesh 0.26 m/s console output used to record the converged pressure and mass-flow results.

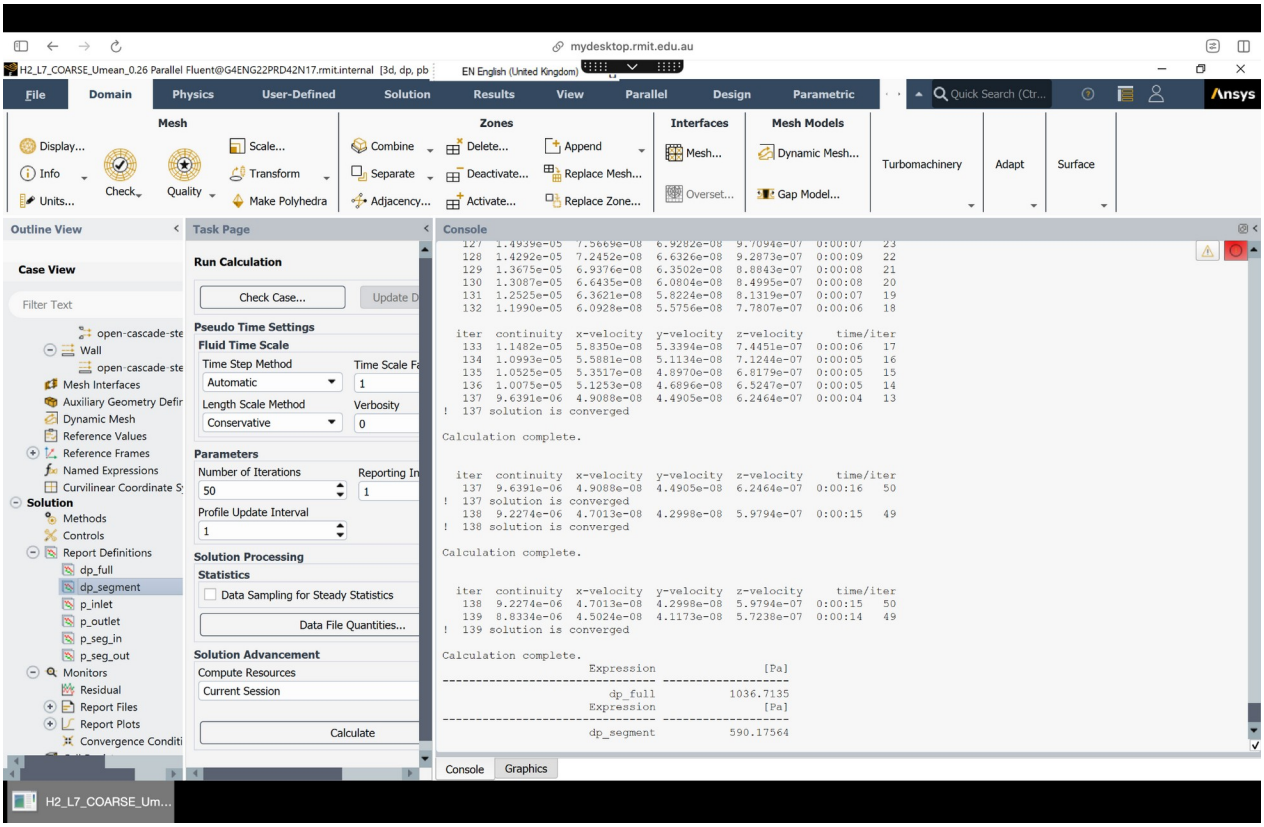


Figure 7. Coarse-mesh 0.26 m/s convergence confirmation after additional iterations; Fluent reports the solution as converged.

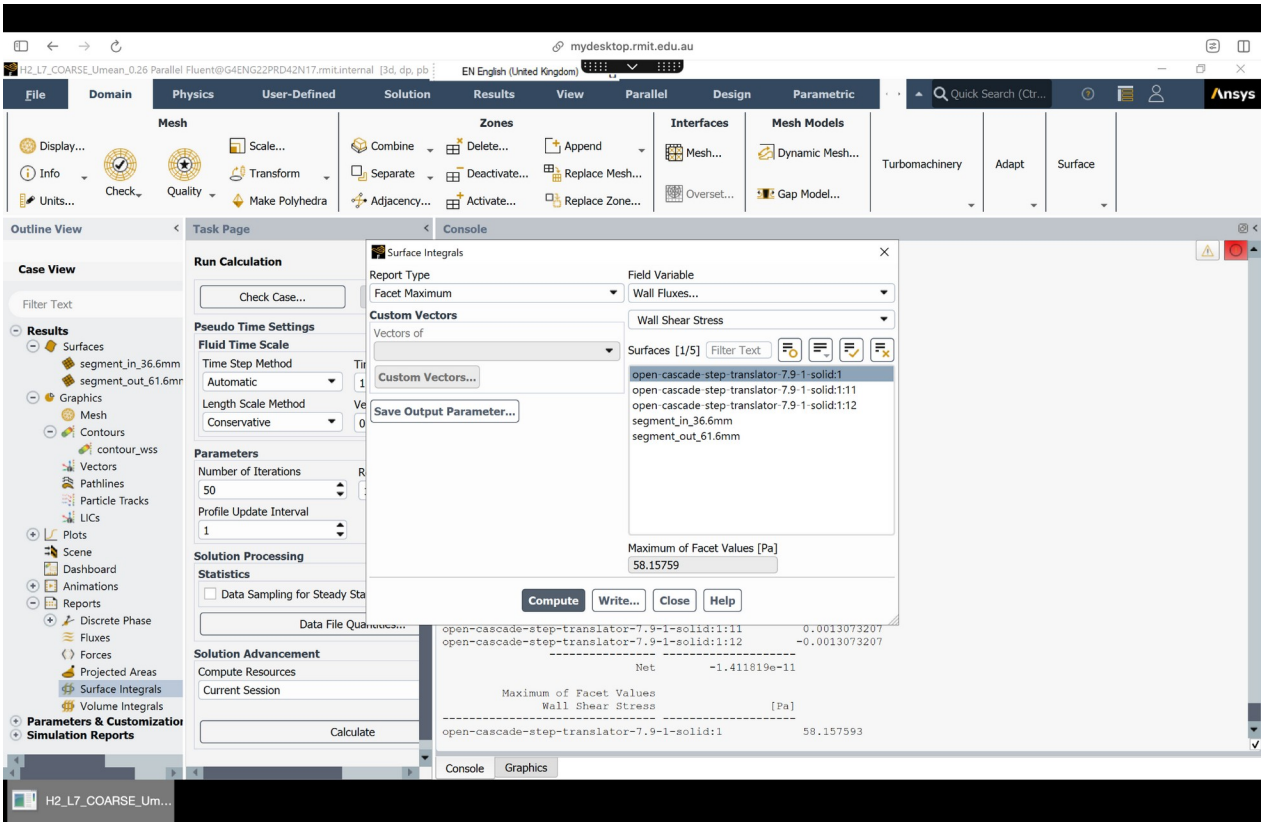


Figure 8. Coarse-mesh 0.26 m/s facet-maximum wall shear stress result.

6. Mesh-Sensitivity / GCI Interpretation

The 25 mm segment pressure drop was the primary ranking quantity. It decreases monotonically with refinement for both inlet velocities:

0.10 m/s: 155.902 → 147.753 → 140.177 Pa (coarse → medium → fine).

0.26 m/s: 590.176 → 578.138 → 561.069 Pa (coarse → medium → fine).

Important verification finding. When the unequal grid ratios are treated using the standard Richardson power-law assumption $\varphi = \varphi_{ext} + C h^p$, the three-grid sequences do not admit a physically meaningful positive apparent order p. The fitted orders are negative (approximately p = -1.13 for 0.10 m/s and p = -2.25 for 0.26 m/s). This indicates that the present three meshes are not demonstrably inside the asymptotic Richardson-convergence range. Therefore, a classical numerical GCI percentage should not be presented as a definitive uncertainty value from these three solutions alone.

This does not invalidate the CFD solutions. It is itself a verification result: the mesh-sensitivity trend is documented, the fine grid provides the highest available resolution, and the pressure/WSS results can be compared transparently. If a formal GCI value is mandatory for the final campaign, the defensible next action is to obtain team approval for either an additional systematically refined grid or a separately justified nominal-order treatment.

7. H2 Completion Status

- Three mesh levels generated and solved.
- Two mean inlet velocities solved on all three meshes (six CFD runs total).
- Convergence checked and, where required, additional iterations performed.
- Full-domain and lesion-segment pressure drops extracted.
- Maximum WSS extracted from the wall using Facet Maximum.
- Mass-flow conservation checked.
- Representative convergence and WSS evidence saved.
- Mesh-sensitivity trend and GCI/asymptotic-range limitation documented.

H2 status: COMPLETE as a CFD mesh-sensitivity/verification study.

8. Files / Evidence to Retain

- Final coarse, medium and fine case/data files for 0.10 m/s.
- Final coarse, medium and fine case/data files for 0.26 m/s.
- Mesh-quality screenshots and cell counts.
- Residual/convergence plots.
- Full-domain and 25 mm segment pressure-drop report outputs.
- Mass-flow report outputs.
- WSS facet-maximum outputs and representative contours.